NVIDIA | AIStore

MLPERF STORAGE V3.0 · CLOSED SUBMISSION

AIStore on Oracle Cloud Infrastructure

Six-node shared object storage with ten benchmark clients

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aistore-6n-oci-bm-denseio-e5-128-10client-retinanet
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SUBMITTER
NVIDIA

SUBMISSION DATE
July 24, 2026

BENCHMARK
MLPerf Storage v3.0
mlpstorage 3.0.40

DIVISION
CLOSED

SOLUTION TYPE
Open-source shared object storage

DEPLOYMENT
Cloud · Kubernetes

POWER REQUIREMENTS
Provider managed; cloud exemption applies

SUBMITTED WORKLOAD
RetinaNet training

10 benchmark clients
BM.DenseIO.E5.128

→

6-node AIStore cluster
6 proxies · 6 targets · 72 target NVMe devices

System topology and measured data path

Clients access AIStore exclusively through its S3-compatible API. A proxy admits each request and redirects the object transfer to the responsible target.

10 × BM.DenseIO.E5.128 benchmark clients
Open MPI · DLIO · s3dlio · equal rank placement per participating host

↓ S3 over HTTP through round-robin across supplied proxy endpoints

6 × AIStore proxy pods
Request admission, cluster map, placement, and HTTP redirect

↓ direct client-to-target redirected object transfer

6 × AIStore target pods on BM.DenseIO.E5.128
72 local NVMe devices · one native distributed object namespace

Architecture reference
For AIStore proxy, target, routing, and deployment details, see the AIStore architecture overview.

Cloud infrastructure inventory

10 BENCHMARK CLIENTS

6 AISTORE STORAGE NODES

72 TARGET NVME DEVICES

Component	Benchmark clients	AIStore storage nodes
Quantity / role	10 external MPI hosts	6 Kubernetes nodes; one proxy and one target per node
Cloud shape	BM.DenseIO.E5.128	BM.DenseIO.E5.128
Compute / network	2 × AMD EPYC 9J14 96-Core Processor; 128 captured cores; 1,511 GB captured memory; 100 Gb/s data networking	2 × AMD EPYC 9J14 96-Core Processor; 128 captured cores; 1,511 GB captured memory; 100 Gb/s data networking
Local media	Present but not used for MLPerf data	12 × 7.68 TB TLC NVME per target; 72 devices total
Platform	Oracle Linux Server 8.10	Oracle Linux Server 8.10

Client-role clarification
The same bare-metal shape was used for client and storage roles. Client-local NVMe was present but not used for benchmark data. All submitted objects were stored by the AIStore targets.

Storage capacity

- 552.96 TB aggregate target-drive nameplate capacity.
- Native AIStore bucket on local target NVMe.
- Mirroring and erasure coding disabled.

Cloud-managed facilities

- Physical switching, racks, cooling, and PSUs are provider managed.
- The MLPerf cloud-deployment power and rack-unit exemption applies.

Reproducible software stack

Layer	Version / configuration used	Reference
AIStore	v4.8, commit `42242b334fd57e993919f04acf3c2011cf2db675`	source
MLPerf Storage	mlpstorage 3.0.40; captured code image(s) `48676b4ff9c4e269d66ca7bc36e16b8e`	19e9f1666e9fe194fb69a6f8f34f53fbe34fe4a2
I/O stack	DLIO with s3dlio; path-style S3 over HTTP	DLIO
Client platform	Oracle Linux Server 8.10	OCI BM.DenseIO.E5.128
Storage platform	Oracle Linux Server 8.10	OCI BM.DenseIO.E5.128

Object path

- Native AIStore bucket through the S3-compatible API.
- Path-style S3 and standard AIStore HTTP redirects.
- Recorded endpoint selection: round-robin across supplied proxy endpoints.

MPI placement

- Equal ranks per participating client.
- `--bind-to none --map-by node`.
- Client-local NVMe outside the data path.

Captured non-default S3 and TCP settings

`S3DLIO_FOLLOW_REDIRECTS=1` `S3DLIO_RANGE_THRESHOLD_MB=1024` `S3DLIO_MULTIPART_THRESHOLD_MB=4096` `S3DLIO_OPERATION_TIMEOUT_SECS=300` `S3DLIO_CONNECT_TIMEOUT_SECS=60` `S3DLIO_MAX_RETRY_ATTEMPTS=10` `S3DLIO_POOL_MAX_IDLE_PER_HOST=64` `AWS_MAX_ATTEMPTS=10` `AWS_RETRY_MODE=standard` `net.ipv4.tcp_tw_reuse=1` `net.ipv4.tcp_retries2=8`

The matching YAML is the authoritative per-host hardware, operating-system, network, environment, sysctl, drive, and capability inventory. Secret values are redacted.

CLOSED workload configuration

This page records the submitted workload parameters and a compact snapshot of the validated result artifacts.

Parameter	Submitted configuration
Model / accelerator	RetinaNet / B200
Scale	370 ranks; 37 per client
Dataset	252,000,000 JPEG objects across 25,200 key-prefix subfolders; approximately 74 TiB
Reader	batch size 24; reader.read_threads=0
Invocation	8 epochs; one warm-up plus five measured runs

Validated result snapshot

Workload	Mean AU	Samples/s	Training I/O
RetinaNet	89.25%	166,384.46	50.04 GiB/s

Interpretation
The mean AU, samples/s, and I/O values are the aggregate values reported by the validated MLPerf artifacts. The corresponding packaged result files remain authoritative.

Primary sources and submission scope

These links connect the tested configuration to product specifications, source code, and protocol behavior.

NVIDIA AIStore documentation
https://docs.nvidia.com/aistore

AIStore architecture and deployment portability
https://docs.nvidia.com/aistore/overview

AIStore S3 compatibility
https://docs.nvidia.com/aistore/s3compat

AIStore open-source repository
https://github.com/NVIDIA/aistore

Cloud instance specification
https://docs.oracle.com/en-us/iaas/Content/Compute/References/computeshapes.htm

MLCommons Storage benchmark and rules
https://github.com/mlcommons/storage

Exact AIStore source revision
42242b334fd57e993919f04acf3c2011cf2db675

Document scope
This PDF describes only `aistore-6n-oci-bm-denseio-e5-128-10client-retinanet` and the result artifacts stored under that system ID. Other client counts, workload configurations, software revisions, and clouds have separate same-name YAML/PDF descriptions.

Machine-readable companion

The same-name YAML is the authoritative structured system inventory.

Validated artifacts

Logs, metadata, resolved configurations, summaries, code-image markers, and aggregate results remain in the matching results directory.

Scaling context and disclosures

The chart compares validated results included in this submission. Green marks the system described in this PDF.

OCI RetinaNet training I/O

Matched CLOSED submissions; 3-node baseline

3 nodes 5 clients	I/O	27.09 GiB/s · 1.00×	
6 nodes 10 clients	I/O	50.04 GiB/s · 1.85×	THIS SYSTEM

RetinaNet stores one sample per approximately 315 KiB JPEG object. Each collective training step issues thousands of individual S3 GETs over HTTP/TCP, so request handling, object lookup, and protocol processing consume a larger share of time than payload transfer and limit throughput scaling. RetinaNet reaches 1.85× the refreshed 3-node training I/O at 6 nodes. Each plotted value comes from a separate validated system submission.

100 Gb/s
nominal per storage node

Conservative network reference
6 storage nodes correspond to **69.85 GiB/s** of aggregate nominal interface bandwidth. Validated training I/O is **50.04 GiB/s**, **71.6%** of that aggregate reference.
This is an advertised-bandwidth reference, not a measured physical ceiling or an application-level guarantee. OCI shape specification.

Important context and disclosures

- This cluster can be pushed closer to its aggregate nominal network line rate. For this submission, we chose settings that kept accelerator utilization (AU) compliant and performance stable across all six runs.
- This submission used HTTP to keep the benchmark path simple and consistent. AIStore supports HTTPS/TLS. With HTTPS, AIStore disables the file-to-socket sendfile fast path (encrypted responses use buffered transfer, and TLS encryption and handshakes add CPU work), so HTTPS performance may differ.
- NVIDIA AIStore is not a managed service offering. It is open source, and its source code, deployment guidance, and deployment recipes are publicly available. These are deployment-specific measurements, not performance guarantees or competitive claims.

Supporting documentation

The following documents provide disclosure, scaling, and deployment context applicable to this NVIDIA AIStore system.

AIStore Disclosure

AIStore Scaling and Cloud Portability